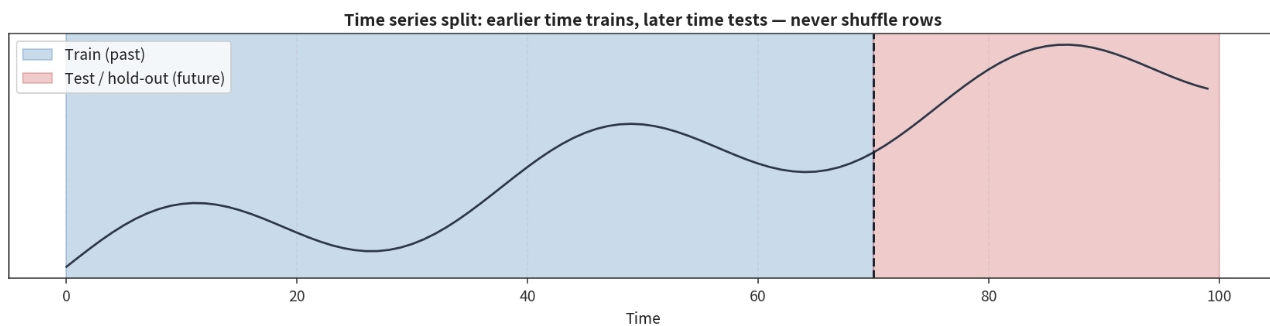


10. Model evaluation metrics

A model can look brilliant on the data it was fitted to and still be useless next month. Evaluation is how you catch that. Two different jobs get mixed up on this slide:

- **Forecast accuracy metrics** — MAE, MSE, RMSE, MAPE — score *predictions against a future they were not allowed to see*.
- **Information criteria** — AIC, BIC — score *how well a model describes the sample it was estimated on*, with a penalty for extra parameters. They help you choose among ARIMA orders. They are not a substitute for a hold-out forecast.

If you remember only one rule: **never shuffle time**. Train on the earlier part, test on the later part.



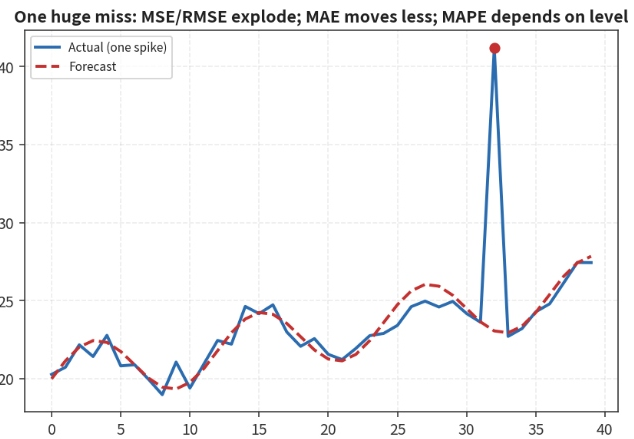
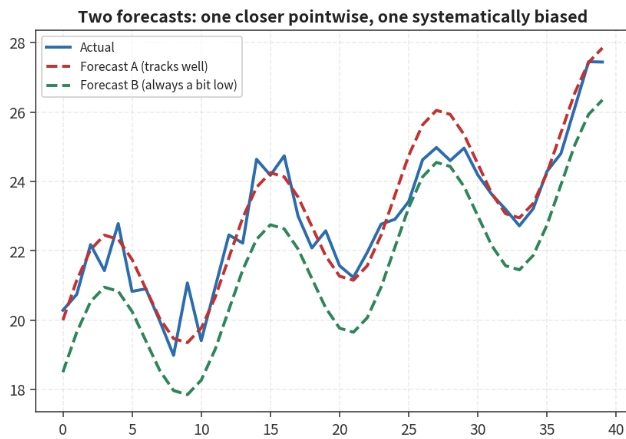
Random cross-validation, the usual machine-learning split, leaks the future into the past. For time series the cut is a date.

10.1 The error you are scoring

At each forecast origin you have an actual value y_t and a prediction $\hat{y}_t$. The error is

$$e_t = y_t - \hat{y}_t$$

All four accuracy metrics are different ways of turning the list $e_1, \dots, e_n$ into one number. They disagree when errors are uneven — especially when one period is a disaster.



Left: two honest-looking forecasts. One tracks; one sits a bit low every period.

Right: the same good forecast, but one actual explodes. MSE and RMSE treat that one miss as a crisis. MAE treats it as one large miss among many. MAPE treats it relative to the *level* of y_t .

10.2 MAE — mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |e_t|$$

What it says: the average size of the miss, in the **same units as the series** (rupees, passengers, degrees).

Properties:

- Every period counts in proportion to how wrong you were. A miss of 10 costs ten times a miss of 1.
- A single monster miss hurts, but it does not dominate the way it does under squaring.
- Easy to explain to a manager: “we are typically off by 120 units.”
- Does not tell you the direction of the bias. A model that is always 5 high and a model that is 5 high half the time and 5 low half the time can share an MAE.

Use MAE when the cost of being wrong is roughly linear and you want a number people can read.

10.3 MSE — mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n e_t^2$$

What it says: average of *squared* misses.

Properties:

- Units are squared (passengers²). That is why people quote RMSE instead.
 - Large errors are punished much more than small ones. Being off by 10 costs 100; being off by 1 costs 1.
 - This matches a world where a big miss is more than ten times as bad as a small miss — stockouts, blown capacity, a bad rate call.
 - One outlier in the test set can make a good model look worse than a dull model that never takes risk.
 - Under Gaussian errors, many estimators that minimise MSE are the familiar least-squares / maximum-likelihood ones. That is why MSE is the mathematical default, not because it is always the business default.
-

10.4 RMSE — root mean squared error

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum_t e_t^2}$$

Same ranking as MSE — any model that wins on MSE wins on RMSE. The square root puts the number **back in the units of the series**, so you can put it next to MAE.

Read the pair together:

- MAE is the typical miss.
- RMSE is pulled up by the tail of misses.

- If RMSE is much larger than MAE, a few periods are doing most of the damage. Look at those periods before you redesign the model.

Rule of thumb, not a law: for errors that are not wild, RMSE is often around $1.25 \times \text{MAE}$. Far above that, hunt outliers or a missed holiday.

10.5 MAPE — mean absolute percentage error

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|$$

What it says: typical miss as a **percent of the actual**.

Why people like it: you can compare a series in rupees with a series in thousands of units. “We are 8% off” travels across products.

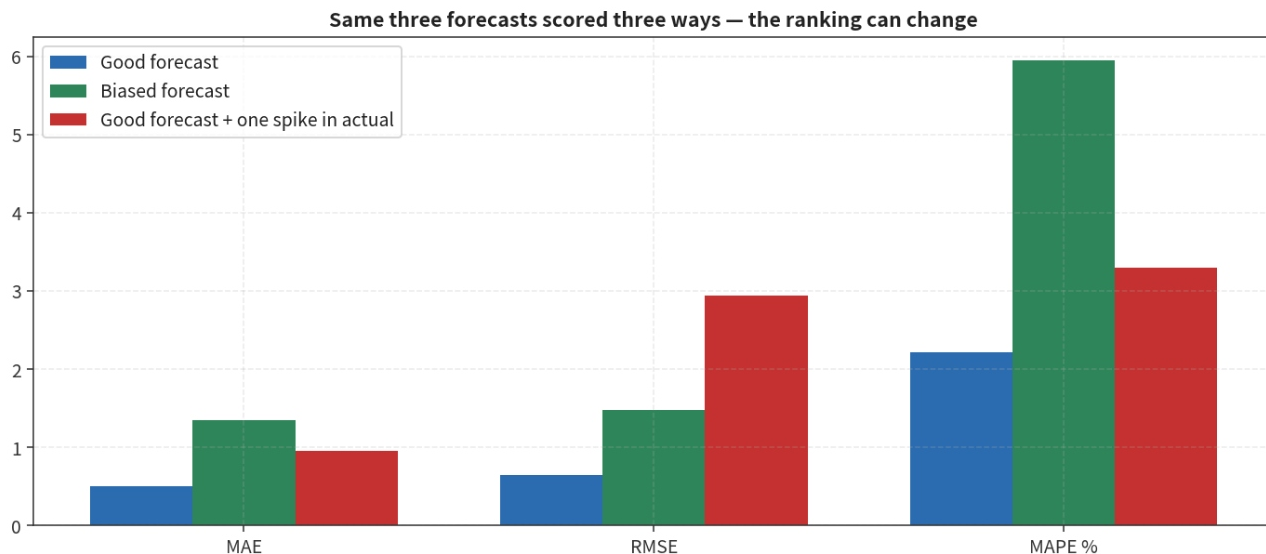
Why it misbehaves:

- **Division by y_t** . If any actual is zero or near zero, MAPE explodes or is undefined. Daily rainfall, counts of rare events, net profit that can be near zero — do not use MAPE.
- It is **asymmetric**. A forecast of 150 when the actual is 100 is a 50% error. A forecast of 100 when the actual is 150 is a 33% error. Same absolute miss, different MAPE. Models that systematically under-forecast can look better on MAPE than they deserve.
- A 10% miss on a small-volume SKU and a 10% miss on the flagship SKU are not the same business cost. MAPE treats them as equal.

Better percentage-style alternatives when you need them later: sMAPE (symmetric MAPE), or MASE (mean absolute scaled error — MAE of your model divided by MAE of a naive seasonal forecast). MASE is what Hyndman recommends when MAPE breaks.

10.6 The ranking can change with the metric

Same three forecasts, three scores:



The “good” forecast wins on MAE. Add one spike in the actuals and RMSE jumps. MAPE depends on whether that spike was a large *percentage* of a small base. If you only ever report one metric, you can crown the wrong model.

A practical convention:

- Report **MAE and RMSE** always, on a hold-out that respects time.
- Add MAPE only if every actual is safely away from zero and percent is how the business talks.
- Look at a plot of e_t over time. A metric will not show you that all the error sits in June.

10.7 AIC and BIC — choosing among models in-sample

Accuracy metrics need a test window. Sometimes you are still in the modelling room, comparing ARIMA(1,1,1) with ARIMA(2,1,0) on the *same* sample. Then you use information criteria.

Both start from the fitted likelihood (or from the residual variance) and add a tax on parameters.

Akaike information criterion

$$\text{AIC} = -2 \log \hat{L} + 2k$$

Bayesian information criterion

$$\text{BIC} = -2 \log \hat{L} + k \log n$$

Here $\hat{L}$ is the maximised likelihood, k is the number of estimated parameters, n is the sample size.

Read the pieces:

- $-2 \log \hat{L}$ gets smaller as the model fits the sample better.
- $2k$ or $k \log n$ gets larger as you add parameters.
- You want the **smallest** AIC or BIC.

BIC's penalty grows with n , so in large samples BIC prefers simpler models than AIC. AIC is a bit more willing to keep an extra lag. Neither is "the truth." They are two different opinions about the same tradeoff.

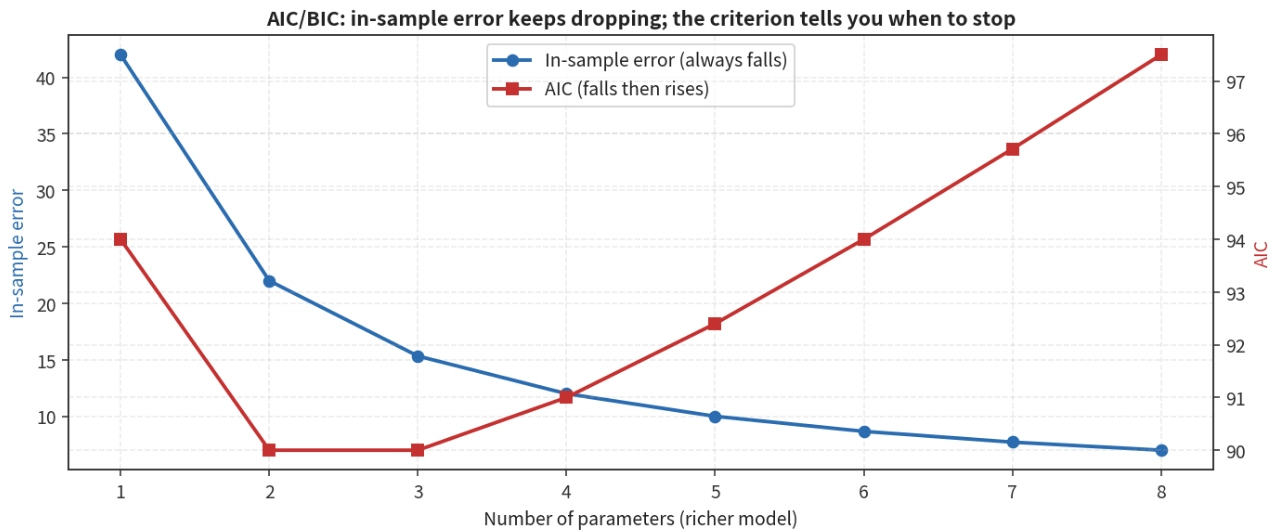
What they are for:

- Comparing ARIMA orders, or SES vs Holt vs Holt–Winters, **on the same series, same transformation, same sample.**
- Auto-ARIMA's search is mostly "try many (p, d, q) , keep the smallest AIC."

What they are not for:

- Comparing a model on raw levels with a model on logs. The likelihood lives on a different scale. You cannot say "the log model has a lower AIC, so it forecasts better."
- Replacing a hold-out RMSE. A model can win AIC and lose next year.

The shape you should have in your head:



In-sample error keeps falling as you add parameters. AIC falls, then rises when the extra parameters are no longer worth their tax. The minimum is the suggested order — then you still check residual ACF and a real forecast window.

A small finite-sample variant, AICc, adds another correction when n is not much larger than k . Use it on short monthly series.

11. Data preprocessing

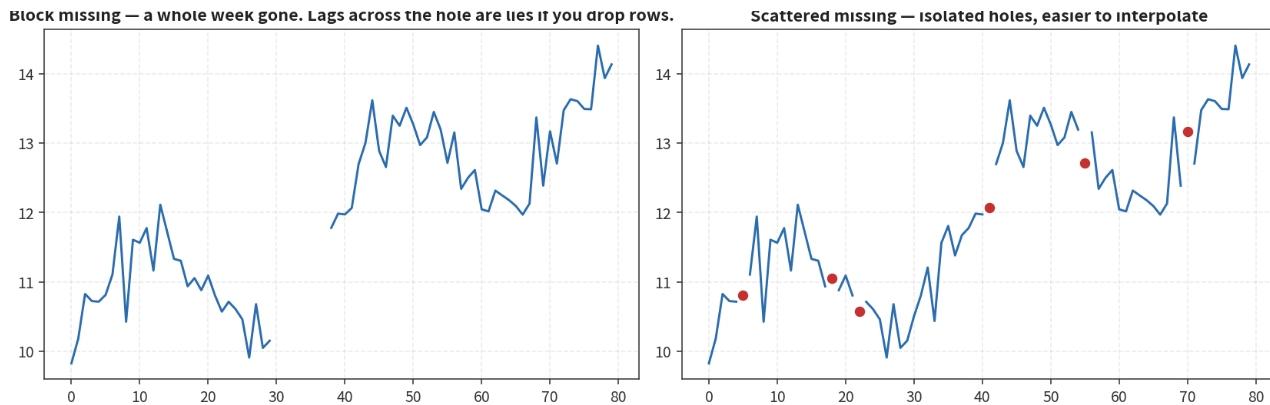
Models assume a regular, reasonably clean series. Real files do not arrive that way. Preprocessing is not a side chore; it changes the lags, the season, and the metrics.

The four bullets on the slide are the usual mess: holes, a wandering level, spikes, and the wrong frequency.

11.1 Missing values

In a cross-section you can drop a row. In a time series, dropping a row **changes every lag**. Yesterday is no longer yesterday. An AR(1) fitted after silently deleting Tuesdays is an AR(1) on a deformed clock.

Two common hole shapes:



- **Scattered holes:** a sensor skip, a Sunday the shop was closed and someone left a blank. Often safe to interpolate if the hole is short relative to the season.
- **Block holes:** a week of missing sales, a month the system was down. Interpolation across a block invents a fake week. Better to mark it, use a model that accepts missing values, or cut the sample.

What people actually do, in rising order of violence:

1. **Keep the timestamp, store NaN.** Do not pretend the day did not exist.
2. **Forward fill** last observation carried forward — acceptable for a slow series and a short hole, deadly if you fill a week of demand with Friday's number.
3. **Linear interpolate** between the neighbours — fine for a temperature gap of two hours, not fine across a holiday week.
4. **Seasonal interpolate** — fill this Tuesday with a blend of nearby Tuesdays. Makes sense when the week is the real unit.
5. **Model-based imputation** (Kalman smoother from a state-space / ARIMA). Best when you already trust a model; circular if you then evaluate that same model on the filled points.
6. **Drop the series until after the hole** if the block is long and early.

Never fill first and then congratulate the model for “predicting” the filled values. Imputed points are not actuals.

Also decide *why* it is missing. A shop closed on Diwali is not the same as a missing upload. The first is a calendar effect; the second is a hole.

11.2 Making data stationary

This is Section 4 applied as a preprocessing step, so only the working recipe here.

You make a series stationary so that AR / MA / ARMA / VAR are allowed to speak. You do **not** always need to: Holt–Winters and many machine-learning setups carry trend and season themselves.

Usual ladder:

1. Plot, rolling mean, rolling std.
2. If variance / seasonal amplitude grows with the level → **log** or **Box–Cox**.
3. ADF + KPSS, with a trend term if the plot climbs.
4. If a unit root remains → **first difference**. Look again.
5. If a yearly wave remains and you are going the ARIMA route → **seasonal difference** Δ_s .
6. Stop as soon as the repaired series has a stable mean and variance and a quickly dying ACF.

Write down every transform. A forecast on differences or on logs must be **inverted** before you compute MAE in the original units. Comparing RMSE on $\Delta \log y_t$ with RMSE on y_t is meaningless.

Over-differencing is a preprocessing error. If ADF already rejects a unit root, do not difference “to be safe.” You will manufacture MA structure and throw away level information.

11.3 Outliers

An outlier is a point that does not belong to the process you want to model: a data typo (extra zero), a one-day flash sale, a strike, a pandemic month.

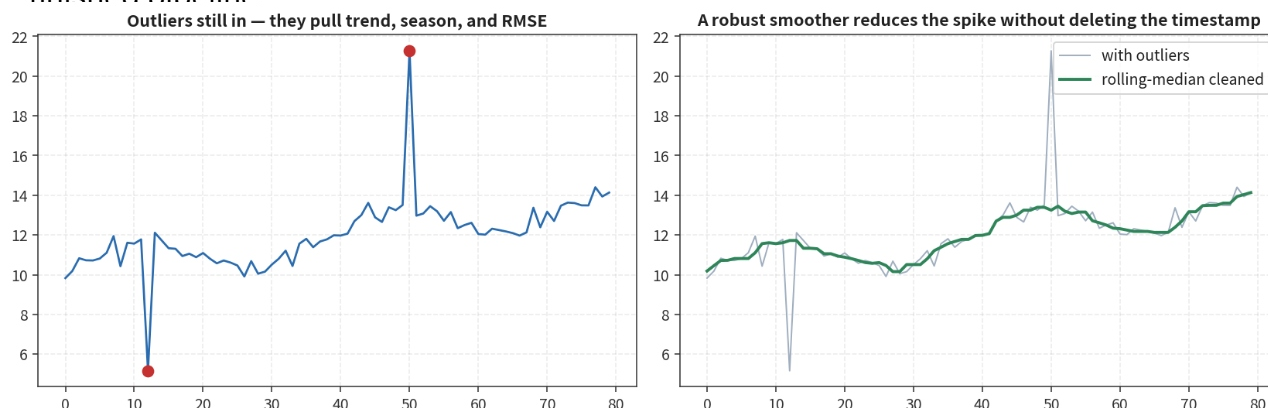
If you leave it in:

- RMSE and MSE will be dominated by that day.
- A moving-average trend will kink.
- Seasonal factors for that month will be pulled.
- An AR coefficient will try to explain a lightning strike as persistence.

If you delete the timestamp, you are back to the missing-value problem.

Better habits:

- **Plot first.** A spike you can name (GST go-live, lockdown) is an intervention, not a statistical mystery.
- **Dummy variables.** Keep the row, add a 0/1 column for that date or that week, let the model estimate a jump. This is the cleanest treatment of a known event.
- **Winsorise or cap** extreme values at a quantile if the spike is measurement noise and you have many of them.
- **Robust decomposition** (STL with robust weights) pushes the spike into the residual so trend and season do not swallow it.
- **Rolling median** as a quick visual clean — see the figure below — is a diagnostic, not a finished pipeline



Do not automatically clip every large residual. Sometimes the spike *is* the thing you were hired to see (a stockout, a fraud burst, a machine failure). Preprocessing should serve the goal from Section 1: description, forecast, explanation, or monitoring.

11.4 Resampling

Resampling is changing the **frequency**: daily → weekly, hourly → daily, monthly → quarterly.

You resample for two honest reasons:

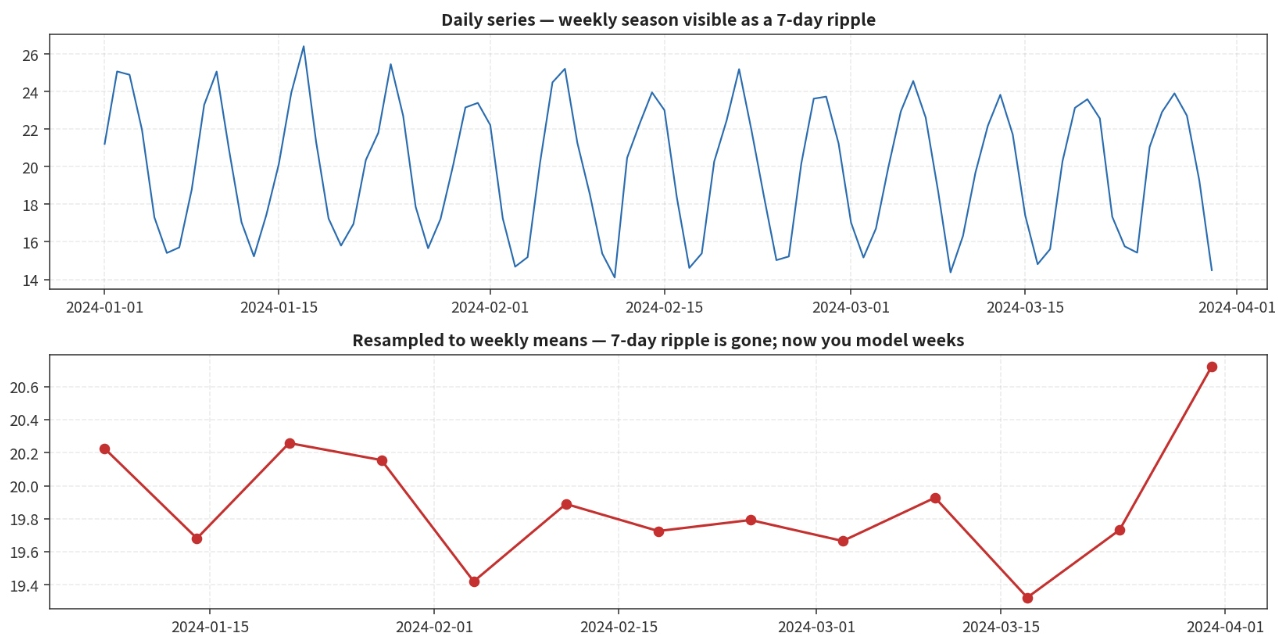
- The decision is made at a coarser clock (you plan inventory weekly, even if sales tick daily).
- The finer season is noise for your purpose (hour-of-day does not matter if you forecast next month).

How you aggregate depends on what the number *means*:

Kind of series	Sensible aggregation
----------------	----------------------

Sales, rainfall, counts, energy used	Sum over the new bucket
Price, temperature, rate, inventory snapshot	Last value, or sometimes mean
A ratio or an already-averaged metric	Think; often weighted mean, not a naive mean of means

Downsampling (fine → coarse) throws information away on purpose. The 7-day ripple in daily data **disappears** when you take weekly means. That can be good (simpler model) or bad (you needed weekday effects).



Upsampling (coarse → fine) invents points you did not observe. Monthly totals spread into fake daily numbers will look like a series and will lie about lags. Avoid it unless you have a real allocation rule.

After you resample, every later choice changes: the seasonal period s , the lags in ACF, the meaning of “one-step forecast.” A one-step forecast on weekly data is next week, not next day. Write the frequency down next to the model.

Other preprocessing that belongs in the same drawer, even though the slide does not list it:

- **Calendar alignment:** time zones, last-Sunday-of-month, 4-4-5 retail calendars.
- **Regularising the index:** make sure there is a row for every expected timestamp, with NaN if needed, before you compute lags.
- **Consistent units:** mix of thousands and units, mix of inclusive tax and exclusive tax — these create fake jumps that ADF will call a unit root.

How 10 and 11 sit on the pipeline

text

raw file

- regular index, missing values, outliers, calendar (Sec 11)
- resample if the decision clock is coarser (Sec 11)
- transform / difference if you need stationarity (Sec 4 and 11)
- fit on the past only (Sec 10 split)
- compare cousins with AIC/BIC (Sec 10)
- score the future with MAE + RMSE, MAPE only if it is legal (Sec 10)
- residual ACF must still look like white noise (Sec 5 and 9)

A model that wins AIC on a series you interpolated through a three-week outage has not been evaluated. A model that wins RMSE on logs has not been evaluated in rupees. Preprocessing and metrics are one conversation.

Resources

- Hyndman & Athanasopoulos, *Forecasting: Principles and Practice* — the chapters on measuring forecast accuracy (MASE is there) and on transformations.
 - Hyndman & Koehler on why MAPE is a poor default.
 - `sklearn.metrics` for MAE/MSE/RMSE; write MAPE yourself and guard zeros.
 - `statsmodels.arma_order_select_ic` / `pmdarima.auto_arima` for AIC search.
 - `pandas.Series.resample`, `interpolate`, `asfreq` for the clock work.
-

That is the full curriculum block from the slides you sent: what a series is, how to take it apart, stationarity, white noise and random walks, the ARIMA family, smoothing, Granger, ACF/PACF, then how to score models and how to clean the file.

If you want a follow-up that ties every section together, the useful one is a **single worked series** (AirPassengers, or a file you paste): missing-value check → STL → ADF/KPSS → difference → ACF/PACF → small SARIMA vs Holt–Winters → MAE/RMSE on a hold-out. Say which series to use